

Internship Project Question

Project Title:

Design and Development of an Inventory Management System for Uganda Industrial Research Institute (UIRI)

Project Question:

Develop a web-based Inventory Management System for Uganda Industrial Research Institute (UIRI) using PHP, MySQL, Laravel, Django, ASP.NET, Node.js, or any other suitable programming framework. The system should enable efficient management of inventory items, tracking of stock movements, generation of reports, and provision of a management dashboard for decision-making.

Project Requirements

The system should include the following functionalities:

1. User Authentication and Authorization

- User registration and login.
- Role-based access control (Administrator, Store Manager, Staff).
- Password encryption and account security.

2. Inventory Management

- Add new inventory items.
- Edit and update item details.
- Delete items when necessary.
- Categorize items (e.g., ICT Equipment, Laboratory Equipment, Office Supplies, Furniture, Machinery, etc.).
- Upload item images.

3. Stock Management

- Record stock received.
- Record stock issued.
- Monitor current stock levels.
- Automatic low-stock alerts.
- Track inventory movement history.

4. Supplier Management

- Register suppliers.
- Update supplier information.
- View supplier transaction history.

5. Reporting Dashboard

The dashboard should provide:

- Total number of inventory items.
- Available stock quantities.
- Low-stock items.
- Recently added items.
- Stock-in and stock-out statistics.
- Graphical reports (charts and graphs).
- Monthly and annual inventory reports.

6. Search and Filtering

- Search items by name, category, supplier, or code.
- Filter reports by date and category.

7. Database Design

Design and implement a relational database containing tables such as:

- Users
- Inventory Items
- Categories
- Suppliers
- Stock Transactions
- Reports

8. Security Features

- Input validation.
- Protection against SQL Injection.
- Session management.
- Audit trail/logging of user activities.

Deliverables

Each student/group should submit:

1. System Proposal.
2. System Requirements Specification (SRS).
3. Database Design (ER Diagram and Data Dictionary).
4. System Design Diagrams (Use Case, Activity, and Class Diagrams).
5. Source Code.
6. Fully Functional Inventory Management System.
7. User Manual.

8. Final Project Report.
9. Presentation and Demonstration.

Assessment Criteria

Component	Marks
Problem Analysis and Requirements Gathering	10
Database Design	15
System Design	15
Implementation and Coding	25
Dashboard and Reporting Features	15
Security and Validation	10
Documentation and Presentation	10
Total	100

Question Provided by ICT Team, Namanve